American Computer Science League

2024-2025 ● Contest 1: Short Problems ● Classroom Division

1. Computer Number Systems

Convert 2024_{10} to octal.

A. 5730
B. 3814
C. 3750
D. 3754
E. 1044

2. Computer Number Systems

What is the value of $9876_{16} + ABC_{16}$ in base 16?

A. 9222
B. 9232
C. A322
D. A332
E. B332

3. Recursive Functions

Find $f(5)$ given:

$$f(x) = \begin{cases} 2 & \text{if } x \leq 1 \\ f(x-1) + x & \text{if } x > 1 \end{cases}$$

A. 11
B. 13
C. 16
D. 20
E. 22

4. Recursive Functions

Find $f(f(f(16)))$ given:

$$f(x) = \begin{cases} f(x-4) + 1 & \text{if } x \geq 10 \\ f(x+1) - 5 & \text{if } 6 < x < 10 \\ 2x & \text{if } x \leq 6 \end{cases}$$

A. 5
B. 8
C. 10
D. 12
E. 13

5. What Does This Program Do? (Branching)

What is the output of the following program?

```
x = 8
y = 9
z = -2
x = y + z
z = y - x
y = y + 1
output x, y, z
```

A. 8, 9, -2
B. 11, 10, 1
C. 7, 8, 2
D. 7, 10, -2
E. 7, 10, 2

6. What Does This Program Do? (Branching)

Run the following program with each of the following input values: 42, 35, 67, 81, 75, 24, 60, 70, 20, 16. How many of them are output?

```
input num
if num % 7 == 0 then
     output num
else
     if num % 5 == 0 then
          output num
     else
          if num >= 60 && num < 80 then
               output num
          end if
     end if
end if
```

A. 8
B. 7
C. 6
D. 5
E. 4

7. Computer Number Systems

Convert 2024_{16} to base 2.

A. 10000001000100
B. 10000000100100
C. 10000000010100
D. 10010010000000
E. 10010000000100

8. Computer Number Systems

Evaluate and express the answer in octal.

$4265_8 + 6524_8$

A. 10701
B. 10789
C. 11001
D. 13011
E. 13110

9. Recursive Functions

Find $f(5)$ given:

$$f(x) = \begin{cases} 256 & \text{if } x \leq 1 \\ \dfrac{f(x-1)}{2} & \text{if } x > 1 \end{cases}$$

A. 128
B. 48
C. 8
D. 100
E. 16

10. Recursive Functions

Find $f(12, 10)$ given:

$$f(x,y) = \begin{cases} f(x-3, y+1) - 2 & \text{if } x > y \\ f(x+1, y-4) + 1 & \text{if } x < y \\ 2x & \text{if } x = y \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 10